CA/Browser Forum Tracker

Part 5 - Revocation and Transparency

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

Revocation and transparency are the two ways the web recovers from a certificate that should not exist. Revocation withdraws trust from a certificate already issued. Transparency makes every issued certificate public, so that a bad one can be found at all.

Both were tested hard in the seven weeks to 19 October. Section 1 covers revocation, where GoDaddy is working through a mass revocation of more than 400,000 certificates and has missed its deadline. Section 2 covers Certificate Transparency, where a wave of new logs qualified and several operators had outages. Section 3 covers Merkle Tree Certificates, a design that folds transparency into issuance itself.

1. Revocation Mechanisms and Timelines

Revocation is the part of the certificate system that has never worked well. The industry's answer has been to route around it with shorter certificates, while the deadlines for revoking a bad certificate stay strict.

Recent Developments

The largest event in this period is a single mass revocation. On 15 October GoDaddy found that 404,607 certificates from its GoDaddy and Starfield hierarchies carried policy information that did not match its own practice statement [1]. Almost all of them are domain validated certificates from the Starfield hierarchy [1]. GoDaddy stopped issuance, corrected the profiles and invoked its mass revocation procedure [1].

The Baseline Requirements gave it five days, to 18 October at 23:45 UTC [2]. GoDaddy said on 16 October that it would not make that deadline and filed the delay as a separate incident [2]. By the deadline it had revoked 170,566 certificates [2]. By early on 19 October the total was 227,422, leaving about 177,000 still valid [2]. GoDaddy gave no completion date and promised its next update for 22 October [2].

Missed deadlines were already a theme in September. The SHA-1 sunset required unexpired subordinate certificates with SHA-1 signatures to be revoked by 15 September [3]. Within three

days, GoDaddy, Firmaprofesional, FNMT and DigiCert each had a delayed revocation report open [3] [4]. Part 4 covers those authorities' wider incident records.

The revocation infrastructure itself drew a wave of problem reports. In the second half of September, more than ten authorities had bugs opened over the content of their revocation lists [4]. They covered missing or forbidden reason codes, expired lists, wrong signature encodings and lists not disclosed in the shared CCADB database [4]. One concerned Let's Encrypt, whose ceremony tool left reason codes off the entries for revoked cross-signed authorities [5].

October brought the same pattern for status responders. Thirteen bugs about faulty online status responses were opened between 8 and 13 October [6]. Several followed problem reports that a third party sent to many authorities at once [6]. The faults included response validity periods, disallowed extensions and responder certificate profiles [6]. Revocation list bugs fell to five in the same window, and no bug of either kind was opened from 15 to 18 October [6] [7].

The country-code registry hijacks of early October showed browser-side revocation at work. Chrome blocked the unauthorized certificates for Google properties through CRLSets at once, then worked with the issuers to revoke them for other clients [8]. It then used transparency log data to find and block certificates for other affected organizations [8]. A community count found 102 certificates from the hijacks, all present in a CRLSet of 10,823 entries [9]. Chrome warned that browser-side blocking may not cover every domain [8].

Let's Encrypt moved its own lifetimes ahead of the Forum's schedule. On 7 October it announced that its default certificates will last 64 days from 10 February 2027, down from 90 [10]. Subscribers may choose 45 or six days instead, and no valid certificate will be revoked in the change [10]. Clients that use the ACME renewal information protocol need no change [10]. Staging switched to 64-day certificates on 14 October [11].

Current State

The online status protocol is being retired rather than repaired. Let's Encrypt ended its responder service in August 2025, citing privacy, because the protocol tells the responder which site a user is visiting [12]. HARICA stopped putting responder addresses in new web certificates in March 2026 and expects to shut down its web responders by 4 May 2027 [13]. The IETF published a lightweight status protocol profile for high-volume environments, RFC 9919, in July 2026 [14].

Short lifetimes are the replacement. Let's Encrypt issued its first six-day certificate in February 2025, and those certificates carry no revocation addresses at all [15]. Renewal is steered by the ACME renewal information protocol, published as RFC 9773 in June 2025, which lets an authority tell clients when to renew [16]. Part 1 covers the Forum's own lifetime schedule.

Browser revocation summaries differ widely in coverage. Firefox's CRLite has covered 99.84% of certificates visible in transparency logs since August 2025, with about 300 kB of daily updates checked locally [17]. Chrome's CRLSets carry about 1% of all revocations, aimed at high-impact and emergency cases [17]. One academic measurement put CRLite's effective coverage during a real incident in November 2025 at 87.8% [18]. Mozilla has posted nothing new on CRLite in this period [19].

Mass revocation now has a written template. Mozilla merged an illustrative plan template on 23 September for the mass revocation plans its policy requires [20]. It defines a mass revocation event as at least 100 certificates or 1% of an authority's active web certificates [21]. Plans must be tested at least every 12 months [21]. HARICA's revocation of 63,525 certificates in July 2026 over a profile error, completed within days, is an earlier example of such an event [22].

The meaning of revocation itself is under discussion. SC-107 would count a certificate as revoked once its serial number appears in a revocation list or draws a revoked status response, from 15 February 2027 [23]. It would also set how quickly an authority must judge a problem report [23]. The ballot was still in discussion on 18 October, past its scheduled end, and Part 1 follows it [23].

Unresolved Conflicts

Two questions sit open. The first is when Mozilla's mass revocation testing became auditable. One Mozilla policy change describes annual tests that revoke 30 random certificates within five days, with audit attestation after 15 April 2026 [24]. Mozilla's template and its earlier guidance date third-party assessment from the audit cycle beginning June 2025 [21] [25]. The current text of the policy's revocation section would settle it.

The second is whether the Forum's revocation deadlines will split. Working group discussions in 2025 proposed separate tracks for breaches of the Baseline Requirements and for deviations from an authority's own policy, with 90 days for the latter [26]. GoDaddy's mismatch with its own practice statement is running on the five-day clock under the current rules [1] [2]. A ballot on the split entering discussion would settle the question.

2. Certificate Transparency

Every publicly trusted web certificate must appear in public logs, and the browsers admit, monitor and retire the logs themselves. The system is midway through a move from its original design to a static, tiled one that can be served from ordinary storage.

Recent Developments

Chrome's log list reached version 95.3 on 17 October [27]. GoDaddy became a log operator when its five Aquamarine tiled logs qualified on 30 September [27]. Sixteen more logs qualified on 13 and 14 October [27]. They are HARICA's Selene logs for 2027 and 2028, Sectigo's Monument and Statue logs, and Let's Encrypt's Sycamore and Willow logs for 2028 [27]. No log in Chrome's list was retired or rejected in the period [27].

Apple's list moves on its own schedule. On 5 October it rejected fifteen shards covering the first half of 2026, as those periods closed [28]. It has made no change since [28].

Operators reported a run of outages. Geomys's Tuscolo log had write delays on 15 and 16 September, and an unpinned software update destabilized IPng Networks' log frontends [29] [30]. GoDaddy traced timeouts on one Aquamarine endpoint to a slow gateway path [31]. Microsec's five logs went down for 42 minutes on 23 September when server access logs filled a disk [32]. Geomys paused submissions to its Trastevere log for about two hours on 14 October during a network fault [33].

The most serious incident is HARICA's. Its new Selene logs missed their maximum merge delay on 12 October and again on 14 October, with six short outages on the second day [34]. HARICA attributed the first miss to unannounced maintenance [34]. Its preliminary report of 15 October does not identify the crash cause, because the server logs were missing and the instance was gone before anyone looked [34].

Monitors raised problems with reading the logs as well. On 17 October a monitor operator reported that three TrustAsia logs return at most 32 entries per request and reject parallel requests without saying when to retry [35]. TrustAsia suggested larger aligned blocks, and the reporter asked how a monitor could learn a log's block size without hard-coding it [35]. Cloudflare archived its Nimbus 2025 log on 13 October and will decommission it in the coming weeks [36].

Current State

Both log designs are accepted, and neither is retired. Chrome admits logs built to the original RFC 6962 design or to the static specification [37] [38]. Since 15 April 2026 a certificate may carry timestamps from static logs alone [39]. Chrome will also stop accepting timestamps delivered through status responses in Chrome 148 [40].

Apple now states its terms for both designs. Its log program page, published 3 September 2026, accepts logs built to version 1.0.0 of the static specification as well as the original design [41]. Both kinds must meet 99% uptime [41]. An original-design log may take up to 24 hours to merge a certificate, while a static log may take at most one minute [41].

Obligations on both sides are tightening. Chrome's root program has required authorities to log every web server precertificate since 15 June 2026 [42]. From 1 November, every newly added log must publish a machine-readable record of its operational state, referenced from its operator's record [37]. Chrome's policy runs each log through the states pending, qualified, usable, read-only, retired and rejected [43].

Two operational questions are open. HARICA's Selene logs remain qualified after their merge delay misses, and Chrome has posted no decision on the incident [27] [34]. On 14 October a participant asked whether other operators will adopt Sunlight's "names tiles" extension, which publishes each entry's subjects and names beside the data tiles [44]. Those tiles are unauthenticated, and the question had no replies [44].

3. Merkle Tree Certificates

A Merkle Tree Certificate replaces the usual chain of signatures with a proof that the certificate appears in a signed public log. Transparency stops being a check after issuance and becomes part of issuance itself.

Recent Developments

The design's specifications moved quickly. Version 06 of the IETF draft appeared on 21 September, with authors from Google, Apple, Cloudflare and Geomys [45]. Version 07 followed on 7 October [45]. The same day, the C2SP project tagged the family of log specifications the

design rests on, with the Merkle certificate log specification at version 0.1.0 [46]. That family covers tiles, mirrors, cosignatures, checkpoints and witnesses [46].

Chrome pinned its testing program to the new text on 16 October [47]. New requests to act as a cosigner must follow version 07 [47]. Chrome keeps version 05 working for a time, drops version 06, and fixes the tiles and mirror specifications at version 0.1.0 [47].

Issuers are building to it. Let’s Encrypt’s issuance software is landing code for Merkle certificate trees, including reading checkpoints from tiles [48]. Cloudflare says its planned authority will run mirroring cosigners and a new family of static transparency logs [49]. Chrome’s testing program already requires every participant to operate both a log and a mirror [50].

Current State

None of this is yet a production requirement. Chrome plans the production phase of its quantum-resistant program for 2027 [50]. Cloudflare has published nothing newer on Merkle certificates since its 29 September announcement [49]. Part 4 covers the root program policies that will decide which of these authorities browsers trust.

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